

HARISH RAGAVENDRA

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SUMMARY

AI and Data Engineer with a strong background in building machine learning pipelines, scalable data workflows, and cloud solutions, emphasizing Python, PySpark, Databricks, and SQL. Proven expertise in feature engineering, workflow automation, and explainable AI across various domains, with developing familiarity in key GCP services. Seeking to leverage experience in data quality and observability for an Analytics Engineering role.

TECHNICAL SKILLS

Languages: Python, SQL.

AI / ML: Scikit-learn, XGBoost, LangChain, SHAP, feature pipelines, anomaly detection.

Data Engineering: PySpark, Spark, Kafka, Databricks, ETL Pipelines, Hadoop (HDFS), medallion architecture, data quality checks, data observability, Orchestration.

Cloud: AWS (S3, EMR, SageMaker, Lambda, Kinesis), Azure (Databricks, Blob Storage), GCP — Familiar/Learning (BigQuery, Dataform, Cloud Composer, Pub/Sub, Datastream, Cloud Run)

Databases / Tools: MongoDB, Snowflake, MySQL, Oracle, Pinecone, Power BI, Tableau, Git, FastAPI, REST API, Docker

Dev Practices: Version control (Git), CI/CD pipelines, unit testing, linting

PROFESSIONAL EXPERIENCE

Associate Software Engineer / Data Engineer | (Remote)

Mar 2023 - Apr 2026

Tanama Software, Bangalore

- Built data ingestion and transformation pipelines using Python, PySpark, and Azure Databricks, processing data from MongoDB and Azure Blob Storage for analytics and ML use cases.
- Contributed to enterprise data migration from HDFS to cloud storage, restructuring data layers using medallion architecture (bronze, silver, gold) to improve data availability for downstream AI workloads.
- Developed and scheduled Databricks workflows to automate recurring data preparation and ML processing tasks, improving pipeline execution reliability by ~30% and reducing manual effort by ~5 hours/week.
- Implemented data quality checks within ETL pipelines including null handling, schema validation, and row-count reconciliation to ensure downstream reliability for ML workloads.
- Developed ETL pipelines for financial reconciliation and demographic sales reporting with schema validation, surfaced through Power BI dashboards for finance and marketing teams.
- Supported end-to-end deployment of an ML solution for fraudulent website detection, managing inference workflows and data integration across Amazon S3 and AWS EMR.

IoT / AI Model Developer Intern |

Jun 2022 - Nov 2022

Diplomatic Consultancy Services, Oman

- Developed AI-based image analysis models for pest detection and anomaly recognition using sensor and camera data in smart farming environments.
- Built data workflows combining IoT devices, edge inputs, and cloud-connected pipelines to support real-time agricultural monitoring.
- Worked on ESP32-based prototypes using sensors and cameras for automated greenhouse and field monitoring applications, improving model performance to 91%+ detection accuracy in pilot use cases.

PROJECTS

AgroIE — Bilingual AI Farming Assistant | Python · FastAPI · LangChain · AWS · Kubernetes · Next.js

- Built a bilingual RAG chatbot for Irish farmers using 7 FastAPI microservices deployed on AWS EKS, with a Next.js frontend supporting English and Irish (Gaeilge) language detection and response generation
- Implemented a full RAG pipeline using LangChain, sentence-transformers (all-MiniLM-L6-v2), and ChromaDB vector store, ingesting 3 knowledge base documents into 17 chunks across bilingual EN/GA indices with 0.80+ cosine similarity retrieval scores.
- Engineered a language detection service with a two-tier approach (keyword pre-check + langdetect) achieving 99.9% confidence on English and 0.85 confidence on Irish, with AWS Translate integration for EN↔GA translation in production.
- Provisioned cloud infrastructure using Terraform (VPC, EKS, ECR, OpenSearch, S3) and Helm charts for Kubernetes deployment, with a GitHub Actions CI/CD pipeline and Prometheus + Grafana observability stack; local dev stack runs on Ollama llama3.2:1b replacing AWS Bedrock Claude 3 Sonnet at zero cost

Enterprise Transaction Risk Scoring Platform with Explainable AI

- Designed a cloud-based transaction risk scoring pipeline using API Gateway, Kinesis Data Streams, AWS Lambda, SageMaker, S3, and DynamoDB for scalable fraud analysis.
- Built machine learning models to classify financial transactions into risk categories and applied SHAP-based explainability for transparent, audit-focused decision-making.

AI-Driven Greenhouse Automation System

- Developed a machine learning pipeline for greenhouse automation using IoT sensor inputs such as temperature, humidity, and soil moisture.
- Built classification and regression models to support control decisions and actuator behaviour.

EDUCATION

MSc in Computing (Artificial Intelligence) Dublin City University, Dublin, Ireland	Sep 2025 – Aug 2026
B.Tech in Computer Science Engineering (IoT) B. S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India CGPA: 7.3 / 10	Oct 2021 - May 2025